

ECM Fault Indication Not Indicated

Symptoms

- The engine room panel ED3 or remote panel ED3 or gauges does **not** indicate faults because it does **not** have power.
- The engine room panel and remote panel power lamps are illuminated.
- The ECM has active faults.

How To Use This Tree

This symptom tree can be used to troubleshoot fault indication related symptoms. Start by performing Step 1 troubleshooting. Step 2 will ask a series of questions and will provide a list of troubleshooting steps to perform, depending on the symptom.

Shoptalk

The SAE J1939 datalink delivers alarm information, generated by the ECM, to the customer interface box. The customer interface box transfers alarm information to the ED3 on the engine room panel and ED3 or gauges in the remote panel(s).

This tree addresses the ED3 or gauges power supply and return.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check Panel Wiring	
	STEP 1A. Check Engine Room Panel Wiring	ED3 indicate fault(s)?
	STEP 1A-1. Check Engine Room Power Switch Supply Wire (C14 Connector to Control Panel Connector)	Less than 10 ohms resistance?
	STEP 1A-2. Check Engine Room Panel Power Switch Supply Wire (C14 Connector to Instrument Panel Connector)	Less than 10 ohms resistance?

STEPS		SPECIFICATIONS
	STEP 1A-3. Check the Engine Room Panel Return Wire (C14 Connector to Control Panel Connector)	Less than 10 ohms resistance?
	STEP 1A-4. Check Engine Room Panel Return Wire (C14 Connector to Instrument Panel Connector)	Less than 10 ohms resistance?
	STEP 1B. Check Remote Panel Wiring	ED3 or gauges indicate fault(s)?
	STEP 1B-1. Check Remote Panel Power Switch Supply Wire	Less than 10 ohms resistance?
	STEP 1B-2. Check Remote Panel Return Wire	Less than 10 ohms resistance?
STEP 2.	Check Panel System Cables	
	STEP 2A. Check Engine Room Panel Cable	Less than 10 ohms resistance?
	STEP 2B. Check Remote Panel Cable	Less than 10 ohms resistance?
STEP 3.	Check Customer Interface Box Wiring	
	STEP 3A. Check the Engine Room and Remote Power Switch Supply Wire	Less than 10 ohms resistance?
	STEP 3B. Check the Engine Room Panel Return Wire	Less than 10 ohms resistance?

STEP 1. Check Panel Wiring

STEP 1A. Check Engine Room Panel Wiring

Conditions :

- Locate engine room panel

Action	Specification/Repair	Next Step
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Check ED3 display for indication of faults.	ED3 indicate fault(s)? YES	1B
	ED3 indicate fault(s)? NO	1A-1

STEP 1A-1. Check Engine Room Power Switch Supply Wire (C14 Connector to Control Panel Connector)

Conditions :

- Locate engine room panel
- Open engine room panel door.

Action	Specification/Repair	Next Step
Check wires between harness connector and control panel connector. • Disconnect cable C14 from the engine room panel. • Place one test lead on the engine room power switch supply pin on connector C14. • Place the other test lead on the engine room power switch supply pin on the control panel connector.	Less than 10 ohms resistance? YES	1A-2
	Less than 10 ohms resistance? NO Repair : Replace the faulty wire. Refer to Procedure 015-024 (/qs3/pubsys2/xml/en/procedures/115/115-015-024.html).	Repair complete.

STEP 1A-2. Check Engine Room Panel Power Switch Supply Wire (C14 Connector to Instrument Panel Connector)

Conditions :

- Locate engine room panel
- Open engine room panel door.

Action	Specification/Repair	Next Step
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Check engine room panel wiring. • Disconnect cable C14 from the engine room panel. • Place one test lead on the engine room power switch supply pin on the C14 connector. • Place the other test lead on the engine room power switch supply wire at the instrument panel connector.	Less than 10 ohms resistance? YES	1A-3
	Less than 10 ohms resistance? NO Repair : Replace the faulty wire. Refer to Procedure 015-024 (/qs3/pubsys2/xml/en/procedures/115/115-015-024.html).	Repair complete.

STEP 1A-3. Check the Engine Room Panel Return Wire (C14 Connector to Control Panel Connector)

Conditions : • Locate engine room panel • Open engine room panel door.		
Action	Specification/Repair	Next Step
Check wires between harness connector and econtrol panel connector. • Disconnect cable C14 from the engine room panel. • Place one test lead on the engine room panel return pin on the on connector C14. • Place the other test lead on the engine room panel return pin on the control panel connector.	Less than 10 ohms resistance? YES	1A-4
	Less than 10 ohms resistance? NO Repair : Replace the faulty wire. Refer to Procedure 015-024 (/qs3/pubsys2/xml/en/procedures/115/115-015-024.html).	Repair complete.

STEP 1A-4. Check Engine Room Panel Return Wire (C14 Connector to Instrument Panel Connector)

Conditions :

- Locate engine room panel
- Open engine room panel door.

Action	Specification/Repair	Next Step
Check engine room panel wiring. • Disconnect cable C14 from the engine room panel. • Place one test lead on the engine room panel return pin on the C14 connector. • Place the other test lead on the engine room panel return wire instrument panel connector.	Less than 10 ohms resistance? YES	2A
	Less than 10 ohms resistance? NO Repair : Replace the faulty wire. Refer to Procedure 015-024 (/qs3/pubsys2/xml/en/procedures/115/115-015-024.html).	Repair complete.

STEP 1B. Check Remote Panel Wiring

Conditions :

- Locate engine room panel

Action	Specification/Repair	Next Step
Check display for indication of faults.	ED3 or gauges indicate fault(s)? YES	Repair complete.
	ED3 or gauges indicate fault(s)? NO	1B-1

STEP 1B-1. Check Remote Panel Power Switch Supply Wire

Conditions :

- Locate remote panel
- Open remote panel.

Action	Specification/Repair	Next Step
Check remote panel wiring. - Place one test lead on the remote panel power switch supply terminal of the X4 connector. - Place the other test lead on the power switch supply wire at the display.	Less than 10 ohms resistance? YES	1B-2
	Less than 10 ohms resistance? NO Repair : Replace the faulty wire. Refer to Procedure 015-025 (/qs3/pubsys2/xml/en/procedures/115/115-015-025.html).	Repair complete.

STEP 1B-2. Check Remote Panel Return Wire

Conditions : - Locate remote panel - Open remote panel.		
Action	Specification/Repair	Next Step
Check remote panel wiring. - Place one test lead on the remote panel return terminal of the X4 connector. - Place the other test lead on the remote panel return wire at the display.	Less than 10 ohms resistance? YES	2B
	Less than 10 ohms resistance? NO Repair : Replace the faulty wire. Refer to Procedure 015-025 (/qs3/pubsys2/xml/en/procedures/115/115-015-025.html).	Repair complete.

STEP 2. Check Panel System Cables

STEP 2A. Check Engine Room Panel Cable

Conditions :

- Disconnect cable connector C14 from the engine room panel
- Disconnect cable connector C7 from the customer interface box.

Action	Specification/Repair	Next Step
Check the engine room panel cable. • Install a jumper between engine room power switch supply pin and the engine room return pin in connector C14. • Place one test lead in the engine room power switch supply pin in connector C7. • Place the other test lead in the engine room return pin in connector C7.	Less than 10 ohms resistance? YES	3A
	Less than 10 ohms resistance? NO Repair : Replace the cable.	Repair complete.

STEP 2B. Check Remote Panel Cable

Conditions :

- Locate and open customer interface box
- Locate and open remote panel
- Disconnect the remote panel cable from the customer interface box X4 connector.

Action	Specification/Repair	Next Step
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Check the remote panel cable. • Install a jumper between remote panel power switch supply terminal and remote panel return terminal on remote control panel X4 in the remote control panel. • Place one test lead on the remote panel power switch supply terminal of the remote panel cable. • Place the other test lead on the remote panel return terminal of the remote panel cable.	Less than 10 ohms resistance? YES	3A
	Less than 10 ohms resistance? NO Repair : Replace the cable.	Repair complete.

STEP 3. Check Customer Interface Box Wiring

STEP 3A. Check the Engine Room and Remote Power Switch Supply Wire

Conditions :

- Open the customer interface box
- Disconnect engine room cable at connector C7 from the customer interface box.

Action	Specification/Repair	Next Step

Check the engine room power switch supply wire. - Place one test lead on the engine room power switch supply pin in connector C7. - Place the other test lead on the remote panel return pin on the customer interface box terminal strip X4.	Less than 10 ohms resistance? YES	3B
	Less than 10 ohms resistance? NO Repair : Replace the faulty wire. Refer to Procedure 015-023 (/qs3/pubsys2/xml/en/procedures/115/115-015-023.html).	Repair complete.

STEP 3B. Check the Engine Room Panel Return Wire		
Conditions : - Open the customer interface box - Disconnect engine room cable at connector C7 from the customer interface box.		
Action	Specification/Repair	Next Step

Check the engine room panel return wire. • Place one test lead on the engine room panel return pin in connector C7. • Place the other test lead on the remote panel return pin on the customer interface box terminal strip X4.	Less than 10 ohms resistance? YES Repair : Replace the faulty display. Refer to Procedure 015-024 (/qs3/pubsys2/xml/en/procedures/115/115-015-024.html) or 015-025 (/qs3/pubsys2/xml/en/procedures/115/115-015-025.html).	Repair complete.
	Less than 10 ohms resistance? NO Repair : Replace the faulty wire. Refer to Procedure 015-023 (/qs3/pubsys2/xml/en/procedures/115/115-015-023.html).	Repair complete.